1.
(a)State Gauss' theorem and apply it to obtain an expression for the electric field at a point (i) outside and (ii) inside a charged conducting sphere.
b) What is an electric dipole? Calculate the Electric field due to an electric dipole at a point on the perpendicular bisector of the dipole.
c) Two parallel plates of area 20c * m ^ 2 each are separated by a distance of 10 mm. If one plate carries a charge 2 * 10 ^ - 9 * C and the other carries a charge - 2 * 10 ^ - 9 * C calculate the eclectic field in the space between them.

2.
(a) Define (i)Susceptibility (ii)Permeability (iii) Self inductance
b) Two coils are wound on the same iron rod so that the flux generated by one passes through the other also. The primary coils has N_{p} loops on it and when a current of 2A flows through it, the flux in it is 2.5 * 10 ^ - 4 * wb Determine the mutual inductance of the two coils if the secondary coils has N_{s} loops on it.
c) Describe an expression for Gauss' law as applied to a dielectric.

3.
a) Explain the terms current density and drift velocity of electrons inside a conductor.
b) Derive an expression for the energy stored in the electric field of a charged Capacitor.
c) A copper wire of 3m * m ^ 2 Cross-sectional area carries a current of 5A. Find the magnitude of drift velocity for the electrons in the wire.

4.
(a) State and explain Kirchoff's law
b) Apply the Kirchoff's law to a wheatstone bridge and find the conditional for bridge balance.
c) Show that Coulomb's law of electrostatics can be deduced from Gauss' law.

5.

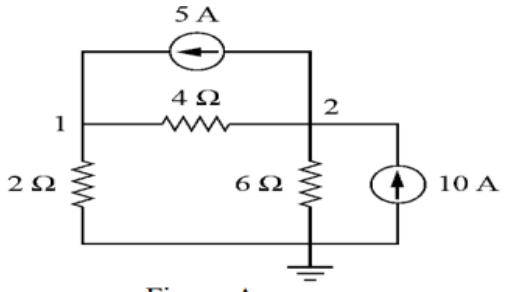


Figure A

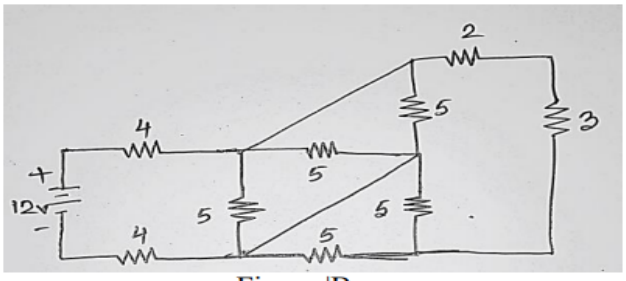
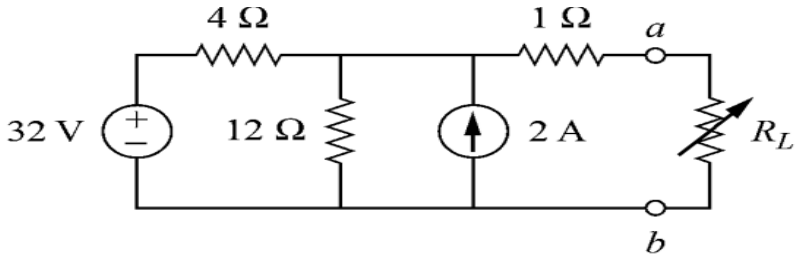


Figure |B

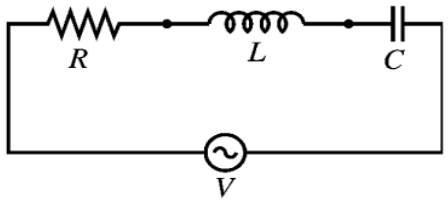
a) State and explain Norton's theorem.
b) Calculate the node voltages in the circuit shown in Figure A
c) Find the equivalent resistance for circuit shown in figure B. Also calculate the return current of the independent source. All resistors are in ohm.

6.



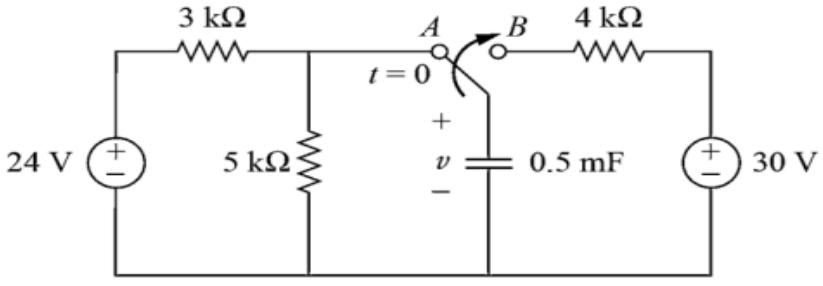
a) State and prove the maximum power transfer theorem for two terminal network.
b) State and explain the Kirchhoff''s current and voltage Laws.
c) Find the Thevenin equivalent circuit of the circuit shown in figure. Then find the current through RL = 6 Ω

7. When, R = 25 Ω, L = 10 mH C = 250 μF V = 230 V, 50 Hz



a) Define instantaneous voltage and RMS voltage.
b) Draw the timing and vector diagram for pure resistive, inductive and capacitive load. What is reactance and impedance of AC circuit?
c) Find Impedance, total current, voltage drop across each elements, draw voltage vector, power absorbed by resistance, power factor from

8.



a) Write down the difference between steady and transient current 2.0
b) Define time constant for suitable example. 3.0
c)The switch in figure has been in position A for long time. At t = 0 , the switch moves to B. Determine the time required for being v(t) = 25 V.